CYBERSECURITY PORTFOLIO | INCIDENT RESPONSE & TOOLING LOG

Incident Handler's Journal: Detection, Analysis & Triage

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Core Focus: Incident Response Lifecycle, Packet Analysis, Threat Intel	**Primary Tools:** Wireshark, tcpdump, VirusTotal, NIST CSF

1. Purpose & Journal Overview

The Incident Handler's Journal is a structured, operational record used by cybersecurity analysts and incident responders to log investigative workflows, track tools utilized across security exercises, and synthesize key takeaways. This journal chronicles hands-on investigations across the core stages of the incident handling lifecycle—from identifying ransomware and phishing attacks, to capturing and inspecting network packets with tcpdump and Wireshark, to triaging indicators of compromise (IoCs) with VirusTotal.

Journal Entry #1 — Date: July 23, 2024

Description	Documenting a Cybersecurity Incident: Healthcare Ransomware Outbreak This incident occurred across two primary phases of the NIST SP 800-61 Rev. 2 incident response lifecycle: 1. Detection and Analysis: The organization initially detected anomalous encryption activity and locked database shares. For technical root-cause analysis, the internal team contacted external third-party digital forensics and incident response (DFIR) specialists. 2. Containment, Eradication, and Recovery: Immediate containment was executed by isolating infected host workstations and shutting down non-critical network segments. External emergency assistance was engaged to validate network eradication and ensure secure, uncompromised backup restoration.
Tool(s) Used	None (Procedural analysis, ticketing systems, and containment protocols).
The 5 W's	• Who: An organized, financially motivated cybercrime syndicate (unethical threat actors). • What: A targeted ransomware security incident resulting in localized file and database encryption. • Where: Enterprise infrastructure and clinical workstations at a healthcare services company. • When: Tuesday at 09:00 AM (peak operational hours). • Why: Initial access was achieved through an employee opening an email attachment via a targeted spear-phishing attack. Once execution succeeded, the payload moved laterally and encrypted critical clinical repositories, leaving a digital ransom note demanding cryptocurrency payment.
Additional Notes & Analysis	1. Prevention Strategy: Deploy multi-layered email security gateways (DMARC, SPF, automated link/attachment sandboxing), enforce strict endpoint execution

	restrictions (disabling macros/PowerShell scripts), and mandate recurring phishing simulations. 2. Ransom Stance: The company should NOT pay the ransom. CISA and FBI guidelines confirm that payment does not guarantee decryption key delivery, encourages criminal syndicates, and often leaves lingering backdoors. The priority must be isolated clean restoration from immutable off-site backups.

Journal Entry #2 — Date: July 25, 2024

Description	Analyzing a Network Packet Capture (.pcap) File Conducted an in-depth packet analysis investigation using graphical network protocol analyzers to identify anomalous application behaviors, inspect protocol handshakes (TCP 3-way handshake), and detect signs of data exfiltration or cleartext credential leakage.
Tool(s) Used	Wireshark (Graphical Network Protocol Analyzer) Value in Cybersecurity: Wireshark captures live network traffic across interfaces and translates raw binary packets into human-readable protocol hierarchies. It empowers security analysts to examine packet headers, filter by IP addresses/ports, inspect payloads, and pinpoint indicators of compromise (IoCs).
The 5 W's	• Who: Security Analyst examining captured enterprise telemetry. • What: Packet capture (.pcap) inspection of HTTP, DNS, and TCP communication streams. • Where: Perimeter boundary interface between the internal LAN and external WAN. • When: Retrospective log and packet review during investigative drills. • Why: To isolate abnormal communication patterns, high-frequency beaconing, and protocol anomalies indicating unauthorized traffic.
Additional Notes & Analysis	Key Observations: While the initial graphical interface with hundreds of packet rows and nested protocol trees felt overwhelming, mastering display filters (such as 'http', 'dns', 'tcp.flags.syn == 1 && tcp.flags.ack == 0', and 'ip.addr == x.x.x.x') made isolating relevant packets rapid and manageable. Wireshark is invaluable for dissecting malicious payloads and confirming whether encrypted tunnels (TLS) are properly enforced.

Journal Entry #3 — Date: July 25, 2024

Description	Capturing and Filtering Packets via the Command-Line Interface (CLI) Executed live packet capture and command-line filtering exercises on headless Linux environments. Filtered traffic based on protocol, port, and host addressing to monitor real-time network interactions directly from the terminal.
Tool(s) Used	tcpdump (Command-Line Network Packet Analyzer)

	Value in Cybersecurity: tcpdump is a lightweight, high-performance packet analyzer natively available on Linux/Unix systems without requiring a GUI. It is indispensable for incident responders performing live triage on remote servers, cloud instances, and restricted terminal-only bastions.
The 5 W's	• Who: Security Analyst configuring interface packet sniffers. • What: Real-time packet capture, filtering, and export into standard .pcap format. • Where: Local host interface (eth0 / lo) on a Linux server. • When: Continuous command-line packet inspection exercise. • Why: To capture targeted diagnostic data with minimal CPU and memory overhead during potential denial-of-service or port scanning events.
Additional Notes & Analysis	Command-Line Mastery & Learning Curve: Navigating tcpdump syntax (e.g., 'tcpdump -i eth0 -nn -c 50 port 80 -w capture.pcap') initially posed challenges due to strict flag syntax. Once understood, the power of Berkley Packet Filters (BPF) became clear—enabling targeted capture without the system bloat of a GUI.

Journal Entry #4 — Date: July 27, 2024

Description	Investigating a Suspicious File Hash & Threat Intelligence Triage Executed an indicator of compromise (IoC) threat lookup following an automated SOC intrusion detection alert. This investigation occurred strictly in the Detection and Analysis phase of the incident response lifecycle. As a SOC analyst, I analyzed a suspicious file hash to distinguish false positives from authentic trojan/malware infections.
Tool(s) Used	VirusTotal & Threat Intelligence Feeds Value in Cybersecurity: VirusTotal aggregates over 70 anti-virus scanners and threat intelligence databases. It allows security analysts to check file hashes (MD5, SHA-1, SHA-256), domains, IPs, and URLs against global threat reputation engines to rapidly determine maliciousness without detonating live payloads in uncontained environments.
The 5 W's	• Who: An unknown external threat actor transmitting malicious payloads. • What: A phishing email containing an infected attachment with the SHA-256 hash: 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b. • Where: An employee endpoint workstation at a financial services company. • When: 01:20 PM, immediately triggering an alert within the Security Operations Center (SOC) from the IDS/EDR. • Why: The threat actor attempted to deliver a trojanized downloader to establish persistence and compromise financial credentials.
Additional Notes & Analysis	Remediation & Prevention: The investigated hash showed widespread vendor detections confirming it as malicious malware. Immediate actions: Blacklist the SHA-256 hash in Endpoint Detection & Response (EDR) agents across all company laptops, block the sender domain on the email gateway, and conduct targeted security awareness training reinforcing email hygiene.

2. Synthesis & Professional Reflections

1. Challenges Encountered & Problem-Solving Approach

Working with tcpdump in the Linux terminal was the most challenging component. Operating in a command-line environment requires exact syntax, proper flag combinations (such as -nn to disable DNS lookups and -v for verbosity), and a solid understanding of Berkeley Packet Filters (BPF). When initial commands produced empty outputs or excessive terminal scrolling, I stepped back, reviewed protocol documentation, and executed targeted tests. This experience reinforced the value of methodical troubleshooting and precision over hurried command entry.

2. Evolution of Understanding in Incident Detection & Response

Before completing this course, incident response seemed like an isolated reactive task—simply stopping malware after it appeared. Throughout this program, my perspective expanded to appreciate the complete lifecycle (NIST SP 800-61 / NIST CSF): Preparation, Detection & Analysis, Containment, Eradication, and Post-Incident Recovery. I recognized that technical tools (SIEM, IDS/IPS, packet analyzers) are only one part of the equation; documented playbooks, cross-functional escalation communication, and continuous post-incident learning are equally vital to enterprise resilience.

3. Most Impactful Tool & Future Technical Direction

Network traffic analysis using Wireshark and tcpdump was the most engaging and valuable area of study. The ability to peer into the microscopic layers of TCP/IP handshakes, inspect DNS query requests, and identify payload anomalies provided a clear understanding of how network attacks occur. Because my primary background lies in Full-Stack Web Development, mastering network protocol analysis bridges application security with underlying network architecture. I intend to deepen these skills through advanced Wireshark packet carving and intrusion analysis drills.